DO NOW — RECALL

Autumn mock feedback and intervention

Name: _____ Class: _____ Date: _____

1. Why must binary search use sorted input?

2. What is a base case?

3. Why do exam questions award method marks?

4. A rising-edge D flip-flop samples D=1 at the first clock edge. D then changes to 0 between edges, and another rising edge occurs. State Q immediately after the first edge, between edges, and after the second edge.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It discards half the candidates based on ordering.

2. Model answer

A condition that stops recursive calls.

3. Model answer

They credit correct intermediate reasoning and algorithm states.

4. Model answer

Q=1 immediately after first rising edge; remains 1 between edges even though D changes; at next rising edge Q becomes 0.

Vocabulary — know these terms

diagnostic — Assessment used to locate a learning gap.

method mark — Credit for a correct intermediate step.

transfer — Applying a method to a changed problem.

correction — Justified replacement for incorrect reasoning.

NEW LEARNING

Read, annotate and explain

Diagnose the weakness

Look for a pattern across the marked questions: omitted intermediate states, incorrect notation or missed explanations.

Rebuild the method

Model each calculation, comparison or state change instead of simply copying the final answer.

Test transfer

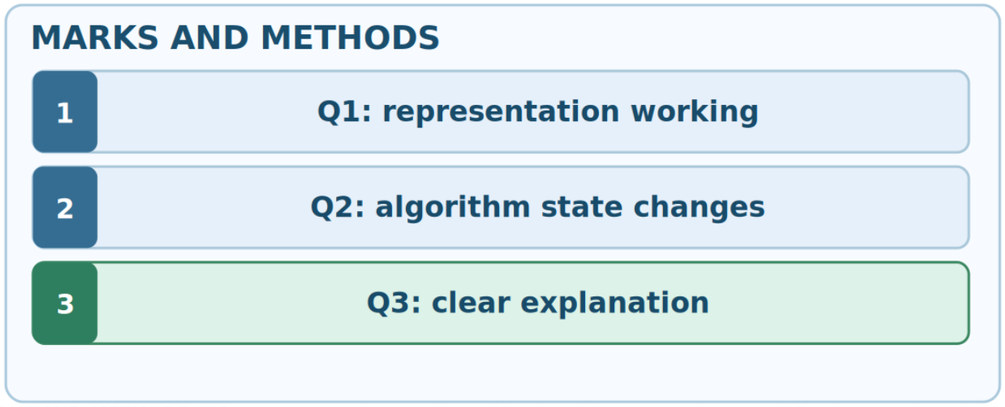
Give students a different array or bit pattern and check whether they can produce the same method independently.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

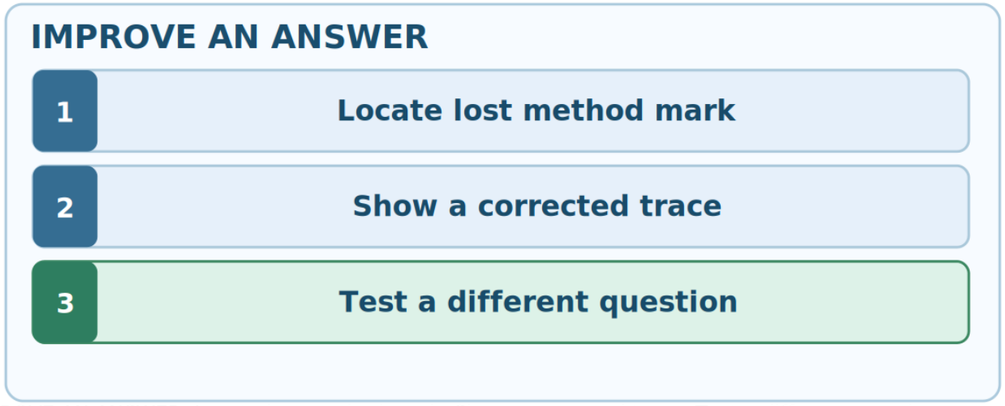
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



MARKS AND METHODS; Q1: representation working; Q2: algorithm state changes; Q3: clear explanation

New Learning



IMPROVE AN ANSWER; Locate lost method mark; Show a corrected trace; Test a different question

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain the answer

The two-edge route costs 6, which is smaller than the direct route cost 9; explain why.

IMPROVE: Explain one correction to your own reasoning.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Copying only a mark-scheme answer proves you can repeat the method.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

I DO — WORKED EXAMPLE

Follow the worked method

Correct an incomplete two's-complement answer for −9 in eight bits.

Predict first

Start with +9 in exactly eight bits and invert it.

Explain the answer

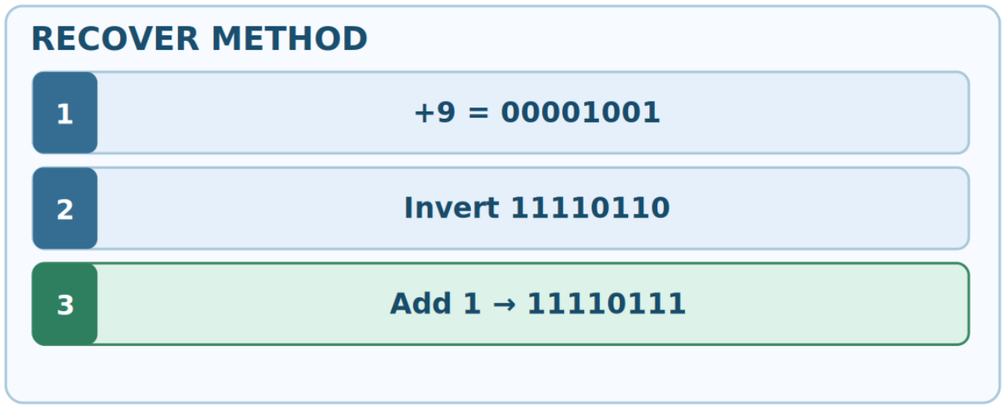
Add one to give 11110111, representing −9. This explicitly shows the stages.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Correct an incomplete two's-complement answer for −9 in eight bits.

Start with +9 in exactly eight bits and invert it.



RECOVER METHOD; +9 = 00001001; Invert 11110110; Add 1 → 11110111

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A student traces binary search for 13 in [1,3,5,7,9,11,13] but does not change high or low after a comparison. Rewrite the trace with bounds and explain why it terminates.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Improve an answer to 'trace linear search for 7 in [3,5,7,9]' by showing comparisons and returned index.

Question 2

Correct the claim '−6 is 11111001 in eight-bit two's complement'. Provide the full method.

WE DO — GUIDED PRACTICE

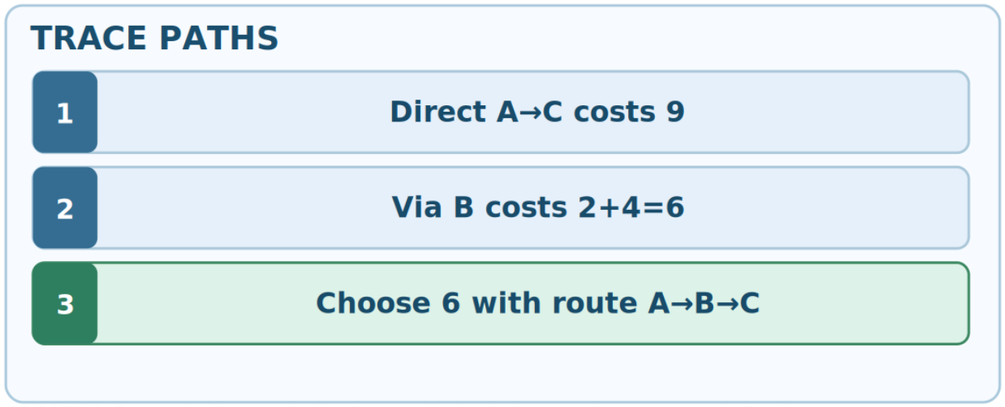
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A student writes only '6' for shortest distance A→C using A–B=2, B–C=4, A–C=9. Improve it.

Compare both candidate routes, recording their total costs.



TRACE PATHS; Direct A→C costs 9; Via B costs 2+4=6; Choose 6 with route A→B→C

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Compare both candidate routes, recording their total costs.

DISCUSS — Compare the representation or steps you used. Explain one decision and challenge one possible misconception.

IMPROVE — Write one precise correction or stronger explanation in your own answer.